

ABSTRACT

Predicting NCAA March Madness Outcomes

Accurately predicting outcomes in the NCAA Division I basketball tournament presents a unique challenge due to its single elimination structure and the high variability inherent in short series competition. This study is motivated by the limitations of traditional bracket based approaches, which rely on binary selections and fail to capture uncertainty in game outcomes. This project investigates predictive modeling of tournament outcomes through statistical and machine learning approaches, with the goal of generating well calibrated win probabilities for every possible matchup.

Following a comprehensive review of existing literature, three modeling frameworks were constructed using R. These include a logistic regression model trained on Dean Oliver's Four Factors of Basketball Success, an XGBoost ensemble model incorporating tournament seeding, strength of schedule adjusted box score statistics, and a custom Elo rating system, and a Monte Carlo simulation framework that derives win probabilities from KenPom efficiency based point predictions. All models are evaluated using the Brier Score, which penalizes both incorrect and overconfident probability estimates. The findings indicate that ensemble methods incorporating dynamic rating systems provide a competitive edge in forecasting the high variance environment of single elimination tournament play.

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INTRODUCTION

The National Collegiate Athletic Association (NCAA) serves as the premier developmental system for the National Basketball Association (NBA) and the Women's National Basketball Association (WNBA). With thousands of athletes across both divisions arriving from around the world, the influence of the NCAA is extensive. As the basketball world has evolved into a multibillion dollar industry [1], the value of a competitive edge has become substantial. Teams historically relied on live scouting and film study to gain these advantages. However, the last decade has seen a major shift toward statistical findings. Inspired largely by the movie *Moneyball*, teams now look to analytics to uncover value that the naked eye might miss.

The focal point of this competitive landscape is March Madness. At the conclusion of the regular season, the selection committee invites 68 teams to participate in this tournament. The field includes the champions from all 32 Division I conferences alongside the best remaining teams chosen at large. The committee seeds these participants into four distinct regions with rankings ranging from 1 to 16. The four teams judged to be the strongest are awarded the number 1 seeds while the next tier receives the number 2 seeds. Within each region, the teams compete in a single elimination format where matchups are initially determined by rank. For example, the top seed plays the sixteenth seed. The winner of each region advances to the Final Four to compete for the national championship. Crucially, these contests take place at neutral sites rather than on home courts. This structure creates a predictive challenge where volatility is the norm.

The NCAA Men's Basketball Tournament is defined by its structural unpredictability. The single-elimination format means that even the most statistically dominant teams face elimination from a single off night, and the compressed six-round schedule amplifies the role of momentum, fatigue, and matchup advantages that aggregate metrics cannot fully capture. This variance is well-documented in existing literature. Extensive benchmarking of advanced ranking systems reveals that even the top-performing predictive models hit a ceiling of approximately 75% accuracy, indicating that roughly one in four tournament games ultimately ends in an upset [29]. The consequences for bracket prediction are severe. The difficulty of bracket prediction further highlights this unpredictability. Mathematical projections place the odds of a perfect 63-game bracket anywhere from 1 in 9.2 quintillion for purely random guesses to 1 in 128 billion for educated predictions. Consequently, there is no record of a publicly verified bracket remaining completely intact beyond the Sweet Sixteen [28].

This inherent volatility drives a massive secondary market focused on sports betting. As legal wagering expands globally, March Madness has evolved into one of the largest gambling events in the world, with \$3.1 billion wagered on tournament outcomes in 2025 [2]. To manage this risk, bookmakers generate odds that serve as the industry standard for predictive accuracy. These odds represent a market consensus that attempts to price in every available variable from player injuries to historical trends. However, research demonstrates that these betting lines are not strictly driven by probability. Evidence indicates that sportsbook forecasts frequently overreact

to new information, creating systematic inefficiencies that violate weak-form market efficiency. This suggests that line-setting is heavily influenced by bookmaker profit margins and public betting sentiment rather than pure statistical likelihood [32]. Consequently, developing a model capable of outperforming these market-established baselines represents the gold standard for predictive success.

This project utilizes the Kaggle March Machine Learning Mania competition as its primary experimental framework. Kaggle provides a setting in which participants submit predicted win probabilities for every possible matchup, rather than making binary bracket selections. The competition evaluates all submissions using the Brier Score, a proper scoring rule that measures the mean squared error between predicted probabilities and binary outcomes [21]. Formally, the Brier Score is defined as:

$$(1) \quad BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2$$

where N is the number of matchups, f_t is the predicted win probability, and $o_t \in \{0, 1\}$ is the actual outcome. A Brier Score of 0 reflects perfect calibration and a score of 0.25 corresponds to the naive baseline of predicting 0.5 for every game. Table 1 illustrates how the score behaves across a range of prediction scenarios.

The table highlights an important property of the Brier Score: overconfident incorrect predictions are penalized more severely than moderate ones. Predicting a 90% win probability for a team that ultimately loses yields

Predicted	Outcome	Scenario	Brier Score
0.50	Win or Loss	Baseline	0.2500
1.00	Win	Perfectly confident, correct	0.0000
1.00	Loss	Perfectly confident, incorrect	1.0000
0.70	Win	Moderately confident, correct	0.0900
0.70	Loss	Moderately confident, incorrect	0.4900
0.90	Win	Highly confident, correct	0.0100
0.90	Loss	Highly confident, incorrect	0.8100

Table 1. Brier Score Examples

a score of 0.81, far worse than the 0.25 baseline. This property incentivizes well calibrated probability estimates rather than extreme predictions, and directly aligns with the objective of this research. The primary goal of this paper is not to compare models against one another but to develop the most accurately calibrated probability estimates possible for submission to the Kaggle competition. Each model is evaluated and refined with this singular objective in mind, and performance is measured exclusively by Brier Score against actual tournament outcomes.

This focus on probability calibration is well suited to basketball because the sport generates a rich and consistent statistical record across all levels of competition. Both individual and collective performances are captured through a standardized set of metrics. Discrete variables such as points, steals, assists, and rebounds provide a consistent numerical foundation for analysis across different eras and levels of competition. This uniformity allows for the creation of structured datasets where team strength can be quantified and compared through objective performance indicators. By leveraging these patterns, machine learning models can identify underlying trends that govern game outcomes and team efficiency.

A basic assessment often relies on a record of wins and losses to infer the overall quality of a team. While these records serve as strong indicators, they frequently fail to capture critical nuance. Furthermore, the single elimination structure of the tournament ensures that upsets remain a constant possibility regardless of historical dominance. A defining example occurred in 2018 when the top ranked Virginia Cavaliers entered the postseason with 31 wins and only 3 losses. They fell to the lowest ranked UMBC Retrievers by 20 points. Such outcomes demonstrate why simple heuristics fail in this environment and why well-calibrated probability estimates are more valuable than confident binary predictions.

Motivated by this challenge, substantial research has addressed the complexities of March Madness prediction. The majority of this work focuses on engineering feature sets for learning algorithms ranging from linear regression to more complex ensemble methods. While the Kaggle competition provides foundational game statistics, transforming these into effective predictors requires significant effort. The following section details the specific data types provided by the competition to establish this baseline. The subsequent section reviews past strategies and the theory behind the modeling techniques. Finally, the paper details the implementation of three distinct modeling techniques, specifically logistic regression, XGBoost, and a Linear Regression Monte Carlo simulation, each evaluated exclusively by their Brier Score performance against the 2025 tournament outcomes, with the best performing model deployed for the 2026 Kaggle submission.

DATA

The highly quantifiable nature of basketball ensures that statistical data is both abundant and accessible. While the Kaggle competition permits participants to incorporate any external datasets they acquire, the organizers themselves provide a comprehensive archive of 37 distinct tables spanning more than three decades, with all tables separated by gender. The archive is divided into regular season and postseason files, with the postseason referred to throughout this project as the playoffs. This research focuses primarily on the detailed results tables for both the regular season and the playoffs, while also utilizing the tournament seeding tables. Although the broader archive extends further back in time, the detailed game statistics required for this analysis commence in the 2003 season, as earlier seasons contain only compact results without the possession-level box score data needed for feature construction. Each entry in the detailed results tables provides a comprehensive statistical breakdown for both the winning and losing teams in every recorded matchup, with the prefix W denoting statistics for the winning team and the prefix L for the losing team. Table 2 summarizes the specific variables recorded for each team.

While raw box scores capture the foundational events of a game, they often fail to account for nuances such as pace of play and strength of schedule. To address this limitation, this research supplements the Kaggle datasets with advanced metrics from analyst Ken Pomeroy (KenPom), which is incorporated into the Linear Regression Monte carlo model for the men's division. Pomeroy, who is widely accepted as the industry standard for college

Variable	Description
PTS	Points scored
FGM / FGA	Field goals made and attempted
FGM3 / FGA3	Three-point field goals made and attempted
FTM / FTA	Free throws made and attempted
OR / DR	Offensive and defensive rebounds secured
Ast	Assists
TO	Turnovers
Blk	Blocks
PF	Personal fouls

Table 2. Regular Season and Playoff Detailed Result Variables

basketball analysis, normalizes performance by calculating efficiency on a per possession basis, where a possession is defined as the period in which a team controls the ball, ending with a shot attempt, turnover, or score. These ratings are updated daily throughout the regular season and tournament. For the purposes of this study, a snapshot of the rankings was acquired from the day prior to the start of the 2025 tournament. Table 3 describes the primary metrics utilized from this source [10].

Metric	Description
NetRtg	Differential between Offensive and Defensive Rating
ORtg	Points scored per 100 possessions, adjusted for opponent
DRtg	Points allowed per 100 possessions, adjusted for opponent
AdjT	Number of possessions per 40 minutes

Table 3. KenPom Metrics

Because Pomeroy’s ratings are exclusively available for the men’s division, this research incorporates an additional dataset from Bart Torvik for the women’s division. Torvik cites Pomeroy as a significant influence, and the ratings are constructed using a comparable methodology. However, distinct

algorithmic differences exist between the two systems; for further detail regarding these distinctions, the reader is referred to [3]. Table 4 describes the metrics utilized from this source.

Metric	Description
ADJOE	Adjusted offensive efficiency
ADJDE	Adjusted defensive efficiency
AdjT	Adjusted tempo

Table 4. Bart Torvik Metrics (Women’s Division)

One important data consideration is the 2020 NCAA tournament, which was canceled due to the COVID-19 pandemic. While regular season data exists for the 2020 season, no tournament outcomes were recorded, creating a gap in the historical playoff results that spans both the men’s and women’s divisions. This means the 2020 season contributes no tournament training observations to any of the models in this analysis, as there are simply no playoff target variables available for that year. Models that train on regular season data may still incorporate 2020 regular season results, but no model draws on 2020 tournament outcomes since none exist.

Across all data sources utilized in this analysis, including the Kaggle box score files, KenPom efficiency ratings, and Bart Torvik rankings, no missing values were encountered that required imputation or row removal prior to modeling. Every team in the dataset had a complete statistical profile across all variables required for feature construction. Any safeguards implemented within individual model pipelines to filter incomplete observations were precautionary measures rather than responses to observed data quality issues. The completeness of the dataset across all sources is

partly a consequence of the 2003 cutoff, which ensures that every season in the analysis has a full year of regular season results available before the tournament begins. Additionally, season identifiers, team identifiers, and games played counts are administrative columns present in the Kaggle files that carry no predictive signal and are excluded from all model feature matrices across every modeling framework in this analysis.

One significant challenge in integrating KenPom and Bart Torvik data with the Kaggle dataset is the inconsistency in team nomenclature. For instance, a university might be listed as “St. Mary’s” in one dataset and “Saint Mary’s CA” in another. To resolve this, a two stage matching pipeline is applied. In the first stage, team names from the external source are normalized to lowercase and matched exactly against the Kaggle team spellings lookup file. Teams that match exactly are assigned their Kaggle **TeamID** directly. In the second stage, teams that remain unmatched after exact matching are passed through a fuzzy matching procedure using the Jaro-Winkler string distance metric with a maximum distance threshold of 0.3 [30]. For each unmatched team the closest spelling in the Kaggle lookup file is selected, allowing minor variations in punctuation, abbreviation, and spacing to be resolved automatically. In practice all teams in both the men’s and women’s divisions were successfully matched, with no teams remaining unmatched.

LITERATURE REVIEW

Past Strategies

The prediction of NCAA tournament outcomes has been studied extensively in both academic and competitive machine learning contexts. Early work demonstrated that tournament seeding alone is a statistically robust predictor of game outcomes, with logistic regression models using a z-score transformation of seed position producing well-calibrated win probabilities across historical tournament data [31]. Subsequent work developed a combined logistic regression and Markov chain model that has since been benchmarked against nearly 100 ranking systems, establishing logistic regression as the baseline for tournament prediction [29]. This ceiling has been confirmed empirically, with comparative evaluations finding that no machine learning classifier, including decision trees, random forests, naive Bayes, or multi-layer perceptrons, was able to exceed 74% to 75% accuracy when trained on NCAAB match data. Also, this research established that feature construction matters more than model choice, with adjusted offensive and defensive efficiencies and Dean Oliver’s Four Factors consistently ranking as the most predictive attributes regardless of the algorithm applied [37]. These insights have been applied directly to the Kaggle March Machine Learning Mania competition, with early entries constructing ensembles of regression functions trained on statistical differentials including ranking index differences, field goal percentage differences, and point differential, demonstrating that fundamental analysis approaches could finish competitively against more complex submissions [35].

The competitive strategies employed by top finishers in the Kaggle competition consistently fall into one of two frameworks that build on this academic foundation. The first relies on fundamental analysis, where the objective is to construct a "true" probability independent of public sentiment. A common implementation involves heuristics such as the Elo rating system, which provides a dynamic measure of team strength that updates after every game and accounts for strength of schedule more effectively than raw records [13]. Many strategies in this framework also incorporate expert efficiency ratings such as KenPom, Nate Silver's FiveThirtyEight ratings, and Massey power ratings, which aggregate performance into a single numeric value. By treating these expert outputs as features rather than final predictions, modelers can leverage domain expertise while correcting for known biases in any individual system.

The second dominant framework relies on the Efficient Market Hypothesis [9]. This approach theorizes that global betting markets, specifically closing lines from major bookmakers, offer a more accurate aggregate predictor of game outcomes than individual statistical models. These strategies do not attempt to beat the market but instead focus on mathematically disentangling the true win probability from the bookmaker's overround and inherent biases. The standard tool in this domain is the `goto_conversion` algorithm [12], which addresses the favorite-long shot bias, the tendency for markets to overvalue underdogs and undervalue heavy favorites, by applying correction functions to raw moneyline prices. By stripping these distortions from closing lines, competitors effectively leverage

the crowd wisdom of the market while removing its pricing inefficiencies. For more information on how this strategy works, we refer the reader to [12].

The applicability of these frameworks is heavily constrained by the structural discontinuity introduced by the 2021 Name, Image, and Likeness (NIL) legislation. An analysis of historical team winning percentages in the post-NIL landscape reveals a significant deviation from historical norms, characterized by a rapid consolidation of talent where the number of programs maintaining elite win percentages above 75% has nearly tripled compared to the preceding two decades [6]. This phenomenon implies that feature weights derived from pre-2021 eras are likely subject to significant concept drift. Consequently, this review restricts its focus to winning strategies from recent Kaggle competitions spanning the last two years, as methodologies developed before the NIL era may not reflect the approaches best suited to the heightened stratification of team strength driven by current transfer portal dynamics.

The most influential baseline in recent competition history is a solution that placed highly in multiple competition years and was directly referenced or built upon by multiple top finishers in both 2024 and 2025 [?]. This solution organizes its feature set into tiers of increasing complexity, beginning with tournament seeding, progressing through season-level efficiency averages, and culminating in a custom Elo rating system and a GLM-based team quality metric derived from point differential. The model is implemented using XGBoost with a fixed number of boosting rounds, and its widespread adoption as a community baseline reflects the consensus that gradient

boosting produces strong and reproducible predictions in this competition. A third consistent pattern across recent competition years is the use of simulation to convert win probability estimates into bracket outcomes. Rather than predicting game results once, top competitors simulate each tournament thousands to hundreds of thousands of times, drawing the winner of each game according to the estimated win probabilities. Among the most notable implementations, one approach simulated brackets using win probabilities derived from expert ratings converted through a normal distribution over predicted point margins, while another used opponent adjusted efficiency margins scaled to a standard possession count with a normal distribution and an empirically estimated standard error. This approach of predicting a continuous score or margin first and then simulating to derive a win probability directly inspired the Linear Regression Monte Carlo model developed in this paper.

The review of these strategies provides motivation for the three models developed in this paper. The logistic regression model is grounded in the academic tradition of sports prediction, serving as an interpretable baseline consistent with the feature classes identified as most predictive in the literature [31, 29, 37]. The XGBoost model is built upon the community baseline that dominated both competition years [?], incorporating seeding, strength of schedule adjusted box score statistics, and a custom Elo rating system with cross-validated hyperparameter tuning. The Linear Regression Monte Carlo model draws on the score prediction and simulation pipeline that proved effective among top 2024 finishers, substituting KenPom and Bart

Torvik efficiency ratings for the external rating sources that top competitors relied upon. Together these three frameworks span the primary methodological strategies that have proven most effective in this competition, grounded in the fundamental analysis approach rather than market probabilities, and evaluated exclusively by their Brier Score performance against actual tournament outcomes.

Modeling Techniques

Having established the empirical context and strategic motivation for the selected frameworks, this section provides the formal mathematical definitions for the algorithms that underpin this paper. While the previous section outlined why these models were chosen, the following details how they function mechanically within the prediction pipeline. Specifically, we define the parameter estimation of Logistic and Linear Regression, the probabilistic simulation mechanics of the Monte Carlo method, the sequential tree construction utilized by XGBoost, and cross validation.

Logistic Regression is a binary classification model that estimates the probability of an outcome by applying the logistic (also known as sigmoid) function to a linear combination of input features. The model is defined as:

$$(2) \quad P(Y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

where Y is the binary outcome, $\mathbf{x} = (x_1, x_2, \dots, x_n)$ represents the feature vector, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_n)$ are the model coefficients estimated through

the maximum likelihood estimation. This function ensures the predicted probabilities are bounded between $[0, 1]$. Logistic regression assumes a linear relationship between the input features and the log-odds of the outcome. This assumption can be restrictive when the true relationships are nonlinear. To address this limitation, common approaches are feature engineering with polynomial or interaction terms (e.g, $x^2, x_i \cdot x_j$) to capture nonlinear patterns.

Linear Regression is a supervised learning method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data. Linear Regression is defined as:

$$(3) \quad \hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

where $\hat{Y}$ is the predicted continuous outcome, $X = (x_1, x_2, \dots, x_k)$ represents the vector of features, and $\beta = (\beta_0, \beta_1, \dots, \beta_k)$ are the model coefficients estimated through ordinary least squares (OLS) minimization. For more information on OLS we refer the reader to [14]. Unlike logistic regression, linear regression produces unbounded predictions and does not inherently output probabilities. In sports prediction contexts, linear regression is commonly used to estimate point totals, scoring margins, or other continuous performance metrics that can subsequently be transformed into win probabilities through additional techniques, such as applying a logistic transformation to the predicted margin or feeding the output into a Monte

Carlo simulation to estimate the probability that a given margin corresponds to a win.

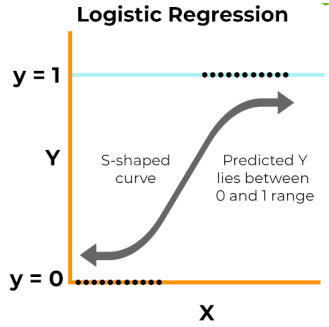


Figure 1. An Example of a Logistic Regression Equation [15].

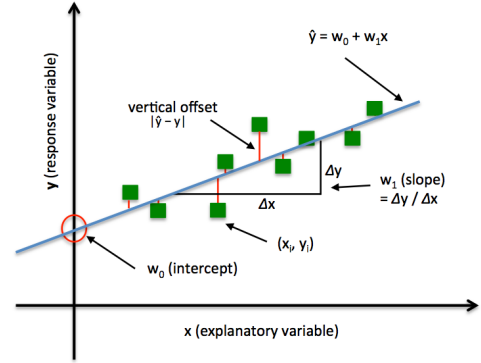


Figure 2. An Example of a Linear Regression Equation [17].

Monte Carlo Simulation is a computational technique that uses repeated random sampling to estimate probabilities for systems with inherent uncertainty. The method generates N independent samples $x_1, x_2, \dots, x_N$ from a probability distribution $f(x)$, then estimates probabilities by computation the proportion of samples satisfying a certain condition. The simulation is defined as:

$$(4) \quad \hat{P} = \frac{1}{N} \sum_{i=1}^N 1(x_i \in A)$$

where $1(\cdot)$ is the indicator function that equals 1 when the condition is satisfied and 0 otherwise. A represents the event of interest, and $\hat{P}$ is the Monte Carlo estimate of the true probability $P(A)$. By the Law of Large Numbers [18], $\hat{P} \rightarrow P(A)$ as $N \rightarrow \infty$. In sports prediction contexts, Monte Carlo methods are frequently employed to simulate game outcomes by

sampling from estimated score distributions, enabling the estimation of win probabilities through empirical frequency counts across simulated matchups.

XGBoost (Extreme Gradient Boosting), considered a "black-box" approach, is an ensemble learning method that constructs a sequence of decision trees where each successive tree is trained to correct the residual errors of the previous trees. Unlike Random Forest, which builds trees independently, XGBoost employs gradient boosting to iteratively minimize a loss function through additive modeling [5]. The model prediction is defined as the sum of predictions from all trees:

$$(5) \quad \hat{y}_i = \sum_{k=1}^K f_k(x_i).$$

Here, $f_k(x_i)$ is the prediction of the k^{th} tree for the i^{th} data point, $\hat{y}_i$ is the final predicted value for the i^{th} data point, and K is the number of trees. Each tree is trained to minimize the objective function

$$(6) \quad \text{obj}(\theta) = \sum_i^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

where l is the loss function which simply computes the difference between the true and predicted, and Ω is the regularization term, discouraging overly complex trees. The optimization process then proceeds via gradient descent on the residuals. For more information regarding gradient descent we refer the reader to [4]. XGBoost is widely adopted in competitive machine learning for

its performance on structured data and ability to capture complex interactions and relationships between features.

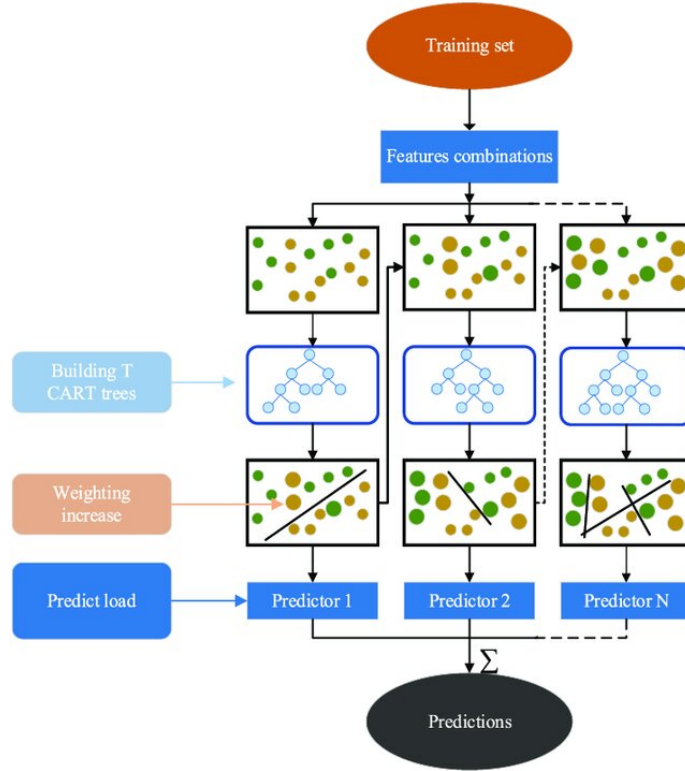


Figure 3. An Illustration Showing How XGBoost Predicts[36].

Cross validation is a resampling technique used to evaluate model performance on data that was not used during training. Rather than evaluating a model on the same data it was trained on, which inherently produces an overly optimistic estimate of performance, cross validation systematically rotates which portion of the data is held out across multiple iterations. This process allows every observation to serve as both a training example and a validation example at different points. In k -fold cross validation, the dataset is partitioned into k equally sized subsets, or folds.

The model is trained k times, each time using $k - 1$ folds as the training set and the remaining single fold as the validation set [27]. The performance metric is recorded for each fold, and the final cross-validated score is computed as the average across all k folds. This methodology produces a significantly more reliable estimate of sampling generalization than a single train-test split because the result is averaged over multiple different held-out subsets rather than depending on the variance of one particular data partition. For more comprehensive information on cross validation methodology, we refer the reader to [26].

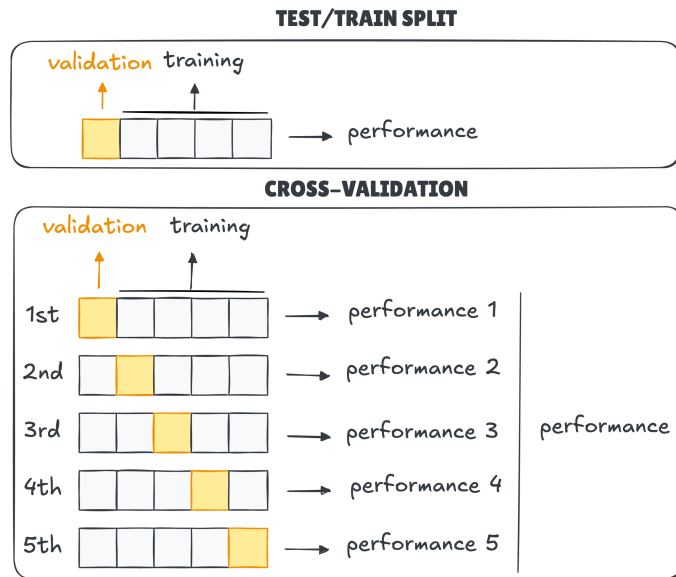


Figure 4. Example of Cross Validation and Regular Test/Train Split [24]

METHODS

The three models developed in this paper span a deliberate range of complexity and methodology, each grounded in the strategies and techniques established in the preceding literature review. Together they cover the primary approaches identified as most effective in this prediction environment: regression-based probability estimation, gradient boosting on structured efficiency features, and simulating win probability. By testing these three frameworks, this research isolates the specific mechanisms that best mitigate the inherent volatility of single-elimination basketball. This design ensures that the theoretical concepts are rigorously validated against multiple analytical philosophies to find the optimal predictive performance.

The simplest approach begins with the premise that a small number of theoretically motivated statistics, computed directly from raw box scores, should be sufficient to capture the fundamental determinants of tournament success. This aligns with the academic finding that feature construction matters more than model complexity, and with the use of logistic regression as the canonical baseline in tournament prediction. The most complex approach abandons that idea entirely, constructing a broad feature matrix and allowing a gradient boosted ensemble to discover relationships across hundreds of variables without prior assumptions about which features matter most, mirroring the community baseline that dominated recent competition years. The third approach occupies a different dimension altogether, bypassing direct classification in favor of predicting each team's expected score and deriving win probabilities through repeated simulation of the

resulting score distributions. Each model is presented in its own subsection following a consistent structure covering variable selection, model implementation, and model analysis. This standardization allows for a direct comparison of their respective strengths and predictive capabilities across the tournament dataset. Each model is evaluated against the 2025 tournament outcomes and benchmarked against the winning Brier Score from that competition year to establish its viability.

Logistic Regression

Logistic regression serves as the natural starting point for this analysis due to its direct alignment with the structure of the prediction problem. The outcome variable is binary, a team either wins or loses, and logistic regression models the log-odds of this outcome as a linear combination of input features, producing a probability as output without requiring additional transformation. Its interpretability is also a key advantage at this stage: the estimated coefficients provide direct insight into which features drive the predicted probability. The independent variables are calculated exclusively from historical regular season data, while the model is trained entirely on historical tournament outcomes. This boundary ensures that the model evaluates how a team's sustained regular season performance translates to the high variance environment of postseason play. To prevent data leakage, the model is trained using a rolling two-year window. For example, to generate forecasts for the 2025 tournament, the model is trained exclusively on tournament matchups from the 2023 and 2024 seasons. This window reflects the structural reality of collegiate athletics, where high roster turnover means

a program undergoes a near-complete personnel transformation within two to three seasons, making older data a poor representation of the current team.

Variable Selection

Analyzing raw box score totals is statistically flawed due to the confounding variable of pace. A team that plays at a high tempo may record high totals for points and rebounds simply because they generate more possessions, not because they are inherently more efficient. To address this limitation, the model utilizes the Four Factors of Basketball Success, developed by Dean Oliver [20], which isolate the four components that correlate most strongly with winning: shooting efficiency, turnover avoidance, rebounding dominance, and free throw frequency. Central to these calculations [16] is the estimation of total possessions ($Poss$), which must be derived from box score data using the following formula:

$$(7) \quad Poss = (FGA - OR) + TO + (0.44 \cdot FTA)$$

The first factor is Effective Field Goal Percentage ($eFG\%$), which adjusts for the added value of three-point shots:

$$(8) \quad eFG\% = \frac{FGM + 0.5 \cdot FGM3}{FGA}$$

The second factor is Turnover Percentage ($TOV\%$), which normalizes turnovers against total possessions:

$$(9) \quad TOV\% = \frac{TO}{Poss}$$

The third factor is Offensive Rebound Percentage ($ORB\%$), which measures rebounding dominance relative to available opportunities:

$$(10) \quad ORB\% = \frac{OR}{OR + DR_{opp}}$$

The fourth factor is Free Throw Rate (FTR), which measures how frequently a team converts free throw attempts relative to field goal attempts:

$$(11) \quad FTR = \frac{FTM}{FGA}$$

Beyond Oliver's original framework, the model incorporates Win Percentage ($Win\%$) as a fifth feature, calculated as the mean of binary game results over the regular season. While the Four Factors capture efficiency across specific phases of the game, Win Percentage serves as a composite signal that aggregates the cumulative effect of all factors including those not directly measured, such as coaching decisions, late game execution, and opponent quality. All five features are computed as season-level averages from

regular season data and then transformed into differentials between the two competing teams prior to modeling, centering the analysis on the relative mismatch between opponents rather than their absolute statistics.

Model Implementation

The logistic regression model is implemented separately for the men's and women's divisions, with the 2025 model trained on tournament outcomes from the 2023 and 2024 seasons. The decision to train independent models rather than a single combined model with a gender indicator reflects the assumption that the Four Factors carry structurally different predictive weight across divisions. Differences in pace, physicality, and three-point volume suggest that the relationship between efficiency differentials and winning probability is not uniform across genders, and forcing shared parameter estimates would risk masking these structural differences. By isolating the training environments, each algorithm is permitted to optimize its coefficients according to the unique statistical footprint of its respective division. The model uses a logit link function to map the linear combination of input differentials to a probability bounded strictly between zero and one. This mathematical transformation is necessary because ordinary linear regression would produce unbounded predictions that cannot be interpreted as viable probabilities. The formal specification for this relationship is defined as:

(12)

$$\begin{aligned} \text{logit}(P) = \ln \left(\frac{P}{1-P} \right) = & \beta_0 + \beta_1(\text{eFG_diff}) + \beta_2(\text{TOV_diff}) \\ & + \beta_3(\text{ORB_diff}) + \beta_4(\text{FTR_diff}) + \beta_5(\text{Win_diff}) \end{aligned}$$

where P represents the probability of a Team 1 victory. The parameter coefficients are derived through maximum likelihood estimation, a process that identifies the β vector that maximizes the probability of observing the recorded tournament outcomes given the input differentials. Specifically, the log-likelihood function takes the following form:

$$\sum_{i=1}^n [y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))]$$

where $y_i \in \{0, 1\}$ is the actual observed outcome for the i -th matchup and $p(x_i)$ is the model's predicted probability of a Team 1 victory [22].

The model relies on three core assumptions. First, the log-odds of the outcome are assumed to be a linear function of the input differentials. Second, all tournament game observations are assumed to be independent of one another. Third, the five input differentials are assumed to exhibit no perfect multicollinearity. The first assumption is reasonable given that the features are already constructed as differentials. The second assumption is standard in tournament prediction contexts where each game is treated as an isolated event. The third assumption is verified through the coefficient analysis in the following section.

A practical constraint of this implementation is the relatively small training sample. A single NCAA tournament produces 63 games per division, yielding approximately 126 combined observations per model iteration. While sufficient for a logistic regression with only five features, this sample size meaningfully limits architectural complexity. Overfitting becomes a nontrivial risk if additional interaction terms or polynomial features are introduced. The model is therefore deliberately kept simple, relying exclusively on the five theoretically motivated differentials. This is consistent with the principle of Occam’s Razor in model selection [23], where a simpler model that generalizes well is preferred over a more complex one that fits the training data too closely.

Predictions for the 2025 tournament are generated by computing the Four Factors directly from each team’s 2025 regular season results. This approach perfectly mirrors the training phase, where every historical tournament game was evaluated strictly against its corresponding regular season efficiency profile. Once compiled, these current metrics are transformed into mathematical differentials for every possible matchup across both the men’s and women’s divisions. The fitted divisional models are then applied via the inverse logit transformation to produce a calibrated win probability, $\hat{P} \in (0, 1)$, for each candidate pairing. Ultimately, this structural consistency ensures that the final forecasts accurately reflect each team’s immediate performance trajectory heading into the postseason, explicitly avoiding the incorporation of outdated statistics from prior academic years.

Model Analysis

With the models estimated on the 2023 and 2024 tournament data, the analysis begins by verifying that the core assumptions of the logistic regression framework are satisfied before interpreting the coefficient estimates. The collinearity assumption is first assessed through a Variance Inflation Factor (VIF) analysis for both the men's and women's models. All five predictors fall below a strict VIF threshold of 3 in both divisions, indicating that the input differentials operate as sufficiently independent statistical signals. No single predictor inflates the variance of any other to a degree that would destabilize the coefficient estimates or undermine the reliability of the subsequent significance tests.

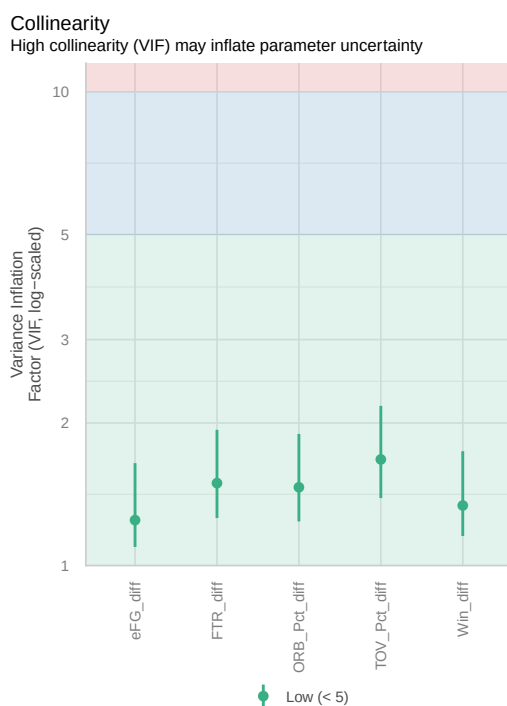


Figure 5. VIF Plot for the Men's Logistic Regression Model.

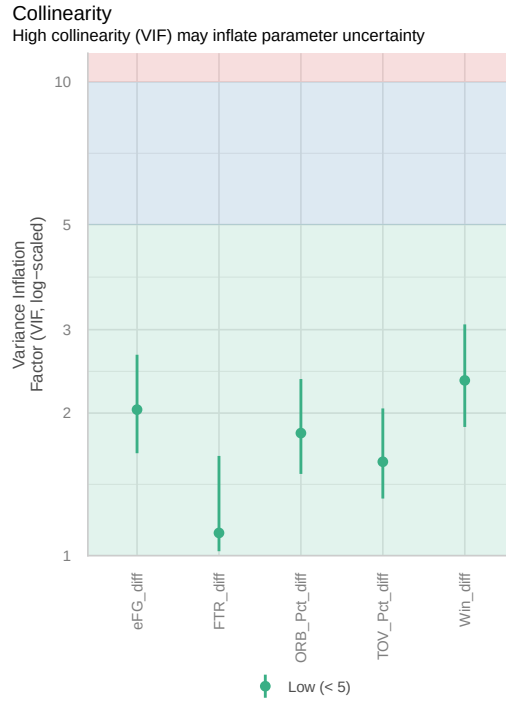


Figure 6. VIF Plot for the Women's Logistic Regression Model.

The pairwise correlation matrices for the men's and women's training data are presented in Figure 7. In the men's model, all pairwise correlations are modest, with the strongest relationship observed between `TOV_Pct_diff` and `FTR_diff` at 0.53, and between `eFG_diff` and `Win_diff` at 0.39. In the women's model, the strongest relationships are between `eFG_diff` and `Win_diff` at 0.57, and between `TOV_Pct_diff` and `Win_diff` at -0.47 . While these moderate correlations are worth noting, none approach a threshold that would indicate problematic multicollinearity, and they are entirely consistent with the VIF results. Together, these two diagnostics confirm that the five differentials can be interpreted individually and that the fundamental collinearity assumption is satisfied in both models.

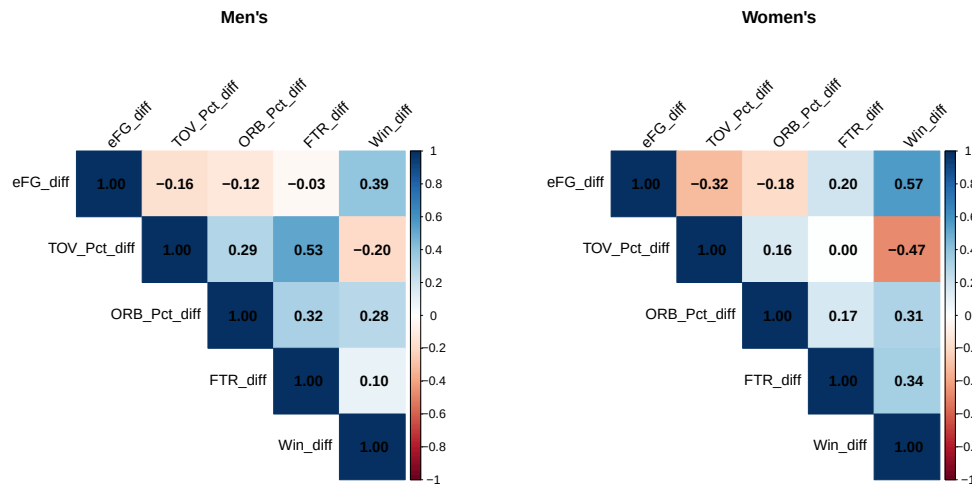


Figure 7. Feature Correlation Matrices for the Men's and Women's Logistic Regression Models.

Because separate models were trained for the men's and women's tournaments, the resulting coefficient structures can be analyzed independently to determine the relative importance of each feature. These independent estimates are then compared against one another to assess whether the Four Factors operate fundamentally differently across the two divisions. By rigorously examining the magnitude and statistical significance of each variable, it becomes possible to identify which specific phases of the game best dictate postseason success for each gender. To illustrate these comparative findings, Tables 5 and 6 present the estimated coefficients, standard errors, p-values, and significance levels for the men's and women's models, respectively.

In the men's model, the only variable to reach conventional statistical significance is Offensive Rebound Percentage, with an estimated coefficient of 12.3634 and $p = 0.0004$. The positive sign is consistent with Oliver's

Variable	Estimate	Std. Error	p-value	Sig.
Intercept	0.4352	0.2062	0.0348	*
eFG Diff	7.7952	6.4160	0.2244	
TOV Pct Diff	-20.5106	12.1028	0.0901	.
ORB Pct Diff	12.3634	4.1219	0.0027	**
FTR Diff	-1.4307	4.5405	0.7527	
Win Diff	1.3253	1.7033	0.4365	

Table 5. Logistic Regression Coefficient Estimates (Men’s 2025 Model)

Variable	Estimate	Std. Error	p-value	Sig.
Intercept	-0.0888	0.2055	0.6657	
eFG Diff	17.4458	5.5413	0.0016	**
TOV Pct Diff	-19.8212	8.7655	0.0237	*
ORB Pct Diff	12.4308	3.5127	0.0004	***
FTR Diff	3.9742	5.1691	0.4420	
Win Diff	-1.8468	2.2764	0.4172	

Table 6. Logistic Regression Coefficient Estimates (Women’s 2025 Model)

framework [20], as a team that dominates the offensive rebounds generates additional possessions while simultaneously denying the opponent transition opportunities. Turnover Percentage approaches significance at the 0.10 level with an estimate of -20.5106 , and the negative direction is theoretically correct, as a team that surrenders possessions more frequently than its opponent is expected to win less often. Effective Field Goal Percentage, Free Throw Rate, and Win Percentage all fail to reach significance in the men’s model, suggesting that within the narrow two-year training window, shooting efficiency and win records carry limited additional explanatory power once rebounding and turnover control are accounted for.

The women’s model tells a markedly different story regarding the underlying drivers of tournament success. Three variables reach statistical

significance: Offensive Rebound Percentage (12.4308, $p < 0.001$), Effective Field Goal Percentage (17.4458, $p = 0.0016$), and Turnover Percentage (-19.8212 , $p = 0.0237$). The near identical magnitude of the Offensive Rebound Percentage estimates across both independently trained models, 12.36 in the men's and 12.43 in the women's, strongly suggests that second chance scoring is a universal determinant of tournament success regardless of division. The most substantive structural difference between the two models is the role of Effective Field Goal Percentage, which is the second most significant predictor in the women's model but statistically indistinguishable from zero in the men's. This divergence suggests that shooting efficiency differentials are more decisive in women's tournament outcomes, where a meaningful gap in $eFG\%$ translates more reliably into a victory than in the men's game, where superior athleticism and individual shot creation can often compensate for overall efficiency deficits. Free Throw Rate fails to reach significance in either model, which is consistent with the relative stability of free throw volume across elite programs at the tournament level.

The significant intercept in the men's model warrants brief discussion. The intercept estimate of 0.4352 represents the log-odds of a Team 1 victory when all five statistical differentials are exactly zero, simulating a matchup between two teams with perfectly identical efficiency profiles. Theoretically, the probability of either team winning in a perfectly symmetric matchup should be exactly 0.5, which corresponds to a log-odds of zero and a statistically insignificant intercept. However, the fact that this intercept is statistically significant at the 0.05 level suggests a slight bias within the

training data, where the lower ID team won more frequently even after controlling for all efficiency metrics. Because Team 1 is strictly defined as the team with the numerically lower Kaggle ID, a static designation originally assigned alphabetically rather than by program prestige or talent, this finding clearly does not reflect a genuine basketball phenomenon. Instead, it is most likely a small sample artifact introduced by the restricted training window and should not be interpreted as a meaningful predictive signal for future tournaments.

The discriminative ability of both models is further assessed through the Area Under the Receiver Operating Characteristic Curve (AUC), visually presented in Figure 8. The AUC formally measures the probability that the model assigns a higher predicted probability to a randomly selected win than to a randomly selected loss. Within this statistical framework, a baseline value of 0.5 indicates no predictive ability, equivalent to random guessing, while a value of 1.0 signifies perfect predictive discrimination. The men's model achieves a training AUC of 0.7056, and the women's model achieves a training AUC of 0.7775. These resulting metrics confirm that both individually trained models exhibit meaningful predictive power on the historical data, successfully separating wins from losses well beyond the threshold of random chance.

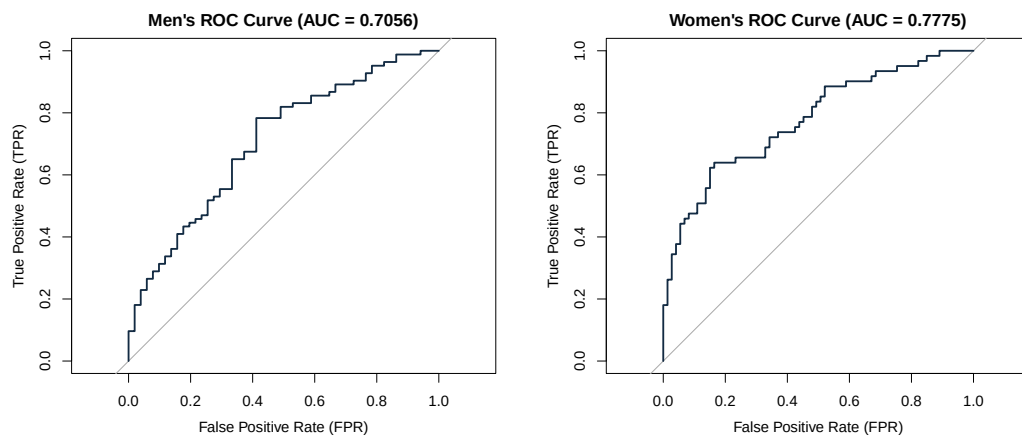


Figure 8. ROC Curves for the Men's and Women's Logistic Regression Models.

Boosting

While logistic regression provides an interpretable and theoretically grounded baseline, its core assumption of a linear relationship between the input differentials and the log-odds of winning imposes meaningful constraints on its predictive ceiling. In reality, the fundamental relationship between team efficiency metrics and tournament success is often highly nonlinear and subject to complex statistical interactions. For instance, a team with a dominant offensive rebounding advantage may only convert that advantage into wins when paired with a sufficient shooting efficiency threshold. This type of conditional relationship is exactly what a linear model cannot capture without explicit feature engineering, prompting the extension of the modeling framework to include XGBoost.

XGBoost is a natural candidate for this problem for several reasons. Unlike traditional parametric models, it does not require strict assumptions about the distributional form of the predictors or their linear relationship to the outcome variable. It is also inherently robust to the scale differences between features, meaning the large magnitude differences observed between metrics in the logistic regression output do not require normalization prior to fitting. Furthermore, XGBoost includes regularization parameters that directly penalize model complexity, which is particularly valuable given the modest total number of tournament observations available across both divisions.

A critical design decision in the data construction phase is the symmetric duplication of each historical tournament game. For every game in

the dataset, two distinct rows are created: one where the winning team is assigned to the Team 1 position, and one where the losing team is assigned to the Team 1 position. The binary outcome variable is then strictly set to 1 when Team 1 wins and 0 when Team 1 loses. This duplication serves three distinct purposes: it allows the model to learn from both the winner's and loser's perspective, it ensures the algorithm does not develop a positional bias toward the Team 1 slot, and it guarantees an exactly balanced target distribution of 50% wins and 50% losses.

The final curated training matrix constructed for this ensemble consisted of 4,552 rows and 139 features, yielding approximately 33 observations per feature. While this ratio would be highly problematic in a traditional parametric setting, XGBoost's rigorous regularization framework and feature subsampling at each tree mitigate this concern by constraining how aggressively the model exploits the feature space. By utilizing these advanced boosting mechanisms, the model can safely navigate a highly dimensional dataset without suffering from immediate overfitting. The following subsections detail the specific variable selection and implementation decisions made for this boosting framework, mirroring the structure of the logistic regression analysis to facilitate a direct comparison between the two approaches.

Variable Selection

A key motivation for the breadth of this feature set is that XGBoost's internal regularization framework absorbs the cost of including additional variables in a way that ordinary regression cannot. In logistic regression,

adding correlated or weakly predictive features inflates standard errors, destabilizes coefficient estimates, and risks overfitting, which is why the Four Factors model was deliberately restricted to five predictors. XGBoost does not share this vulnerability. At each boosting round the algorithm selects only the features and split points that produce the greatest reduction in the loss function, effectively ignoring features that contribute no signal. Features that are redundant are simply passed over during tree construction rather than distorting the model as they would in a linear framework. This property is reinforced by the `colsample_bytree` hyperparameter, which randomly samples a subset of features at each tree, further reducing the influence of any single uninformative predictor. As a result, the XGBoost feature set can be expanded aggressively without the same penalty that would apply in a parametric setting, and the cost of including a feature that turns out to be irrelevant is negligible compared to the cost of omitting one that carries genuine signal.

The XGBoost model is trained on all NCAA tournament games between 2003 and 2024, combining both the men's and women's divisions into a single dataset. This differs from the logistic regression approach, where the divisions are modeled separately. The decision to combine divisions here is deliberate: XGBoost is a high-capacity model with a substantially larger feature set, and merging both divisions increases the number of available training observations. Given that the total number of tournament games in any single division across 22 seasons is modest relative to the complexity of the feature space, this pooling strategy reduces the risk of overfitting and

allows the model to learn more feature relationships. The underlying assumption is that the predictive structure of tournament outcomes, specifically the relationships between efficiency differentials, seeding, and win probability, is sufficiently consistent across divisions to justify training on a combined sample. To ensure the model can still account for any systemic, structural differences between the two tournaments, a binary `men_women` indicator variable is retained in the feature set. This allows the algorithm to learn universal interactions across the pooled data while applying bracket-specific adjustments when necessary.

The feature set for the XGBoost model is organized into three tiers of increasing complexity: seeding features, simple season statistics, and strength of schedule adjusted box score profiles. Together these tiers provide the model with information about committee evaluation, raw performance, and opponent-adjusted efficiency. The first tier consists of tournament seeding. Each team's seed number is included individually as `T1_seed` and `T2_seed`, along with their difference `Seed_diff`, defined as the opponent's seed minus the team's seed such that a positive value indicates a favorable matchup. Including all three seed features rather than the differential alone allows the model to learn nonlinear effects of absolute seeding. A seed differential of one means something very different when it separates a 1 seed from a 2 seed than when it separates an 8 seed from a 9 seed. The lower seeded teams in the tournament are generally far more dominant over their opponents than the differential alone would suggest, and a model that only sees the difference cannot distinguish between these cases. Figure 9 confirms this through two

complementary views. The left panel shows that average point differential declines monotonically as seed number increases across both divisions, while the right panel shows that as the seed differential between two teams widens, the better seeded team wins by larger margins on average, validating seed differential as a strong predictive signal in its own right.

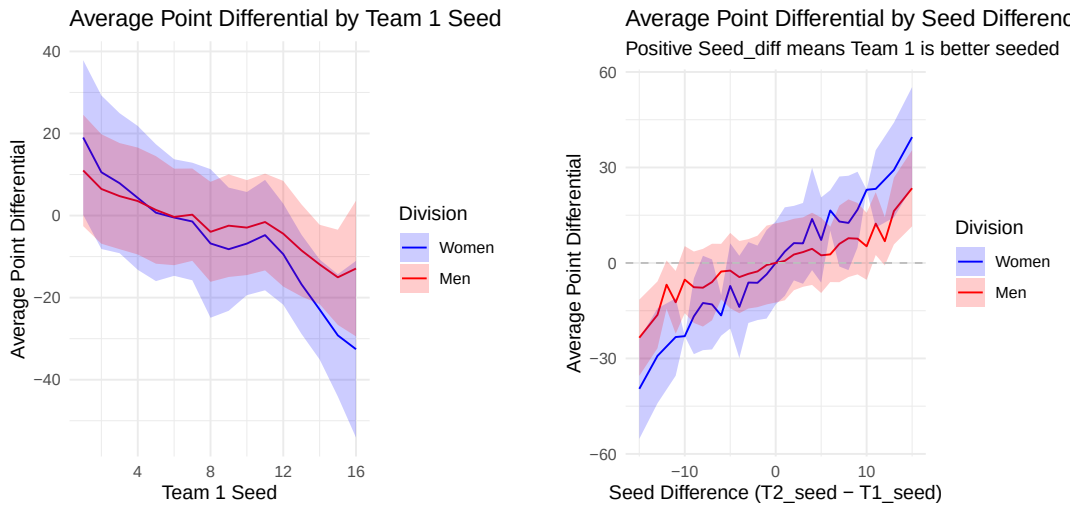


Figure 9. Left: Average point differential by Team 1 seed. Right: Average point differential by seed differential.

The second tier consists of simple season statistics computed from regular season games. For each team these include season win percentage, average points scored, average points allowed, and average point differential. Pairwise differentials between Team 1 and Team 2 are also computed for each of these statistics, providing the model with both absolute team quality signals and direct matchup comparison signals. These features capture overall team strength but do not account for the quality of opposition faced. A team that averaged 85 points per game against a weak conference schedule is not

necessarily more dangerous than a team that averaged 75 points against a gauntlet of top 25 opponents, and the raw averages alone cannot make that distinction. This limitation motivates the third tier.

The third tier introduces strength of schedule adjusted box score profiles. For each of the fourteen raw box score statistics available in the Kaggle data, including field goals made and attempted, three point attempts, free throw attempts, offensive and defensive rebounds, assists, turnovers, steals, blocks, and personal fouls, seasonal averages are computed from two perspectives. The first perspective captures a team’s own offensive and defensive production, prefixed as `Off_` and `Def_` respectively. The second perspective captures the quality of opposition faced by averaging the seasonal profiles of every opponent a team played during the regular season, producing a set of `SOS_` prefixed features. A team whose offensive statistics were produced against consistently strong defensive opponents is credited more heavily than a team whose statistics were inflated by a weak schedule. All box score statistics are adjusted for overtime prior to averaging using the formula $\text{adjOT} = \frac{40+5 \times \text{NumOT}}{40}$, which scales each game’s statistics back to a standard 40 minute equivalent. This ensures that high scoring overtime games do not artificially inflate a team’s seasonal averages.

The final component of the feature set is a custom Elo rating system developed to capture dynamic team strength across the full regular season. Unlike the static season averages described in the previous tiers, Elo ratings are updated after every game and reflect the cumulative result of a team’s performance relative to its opponents throughout the season. This means a

team that starts slowly but peaks heading into the tournament will have a higher Elo rating than its season averages alone would suggest, and conversely a team that fades late in the season will have a deflated rating relative to its win percentage.

The Elo system used here is adapted from the rating system originally developed for chess by Arpad Elo [13], with the implementation structure inspired by [7], originally written in Python and reimplemented in R for this analysis. Each team begins the first season with an initial rating of 1500. After each regular season game the winner gains rating points and the loser loses an equal number, preserving the total rating mass in the system. The magnitude of the exchange depends on how surprising the result was relative to the pre-game expectations implied by the two teams' current ratings. The expected win probability for Team 1 is computed as:

$$(13) \quad E_1 = \frac{1}{1 + 10^{(R_2 - R_1)/400}}$$

where R_1 and R_2 are the current ratings of Team 1 and Team 2 respectively. The denominator of 400 is the width parameter that controls how sensitive the expected probability is to rating differences. After the game resolves, ratings are updated as:

$$R_1 = R_1 + k(1 - E_1)$$

$$R_2 = R_2 + k(0 - E_2)$$

where $k = 64$ is the K-factor controlling the maximum possible rating change per game and $E_2 = 1 - E_1$. When a heavy favorite wins, E_1 is close to 1 and the rating change is small since the result was expected. When a heavy underdog wins, E_1 is close to 0 and the rating change is large. This property makes Elo particularly well suited for tournament prediction because late season upsets and dominant runs cause large rating movements that static averages cannot capture. A key design decision in this implementation is the Elo carry over. Rather than resetting every team to 1500 at the start of each new season, the end of season rating is partially carried forward according to:

$$(14) \quad R_{\text{new}} = 0.5 \times R_{\text{final}} + 750.$$

This means 50% of a team's historical rating is preserved into the next season while the other 50% regresses toward the mean. The motivation is that program quality is not purely year to year. A program like Duke or Connecticut that has been consistently elite for two decades should not start each season at the same rating as an average program making its first tournament appearance. The carry over allows the Elo system to accumulate this historical signal while the regression toward the mean prevents ratings from drifting arbitrarily far from the initial value over many seasons. Figure 10 illustrates the Elo ratings for the top 20 men's and women's programs in 2025, with the left panel displaying the men's rankings and the right panel displaying the women's rankings, reflecting both current season performance and accumulated historical quality. Each team's final Elo rating is included in the feature set as `T1_Elo` and `T2_Elo`, along with their difference `Elo_diff`.

Including all three allows the model to learn both absolute and relative rating effects simultaneously.

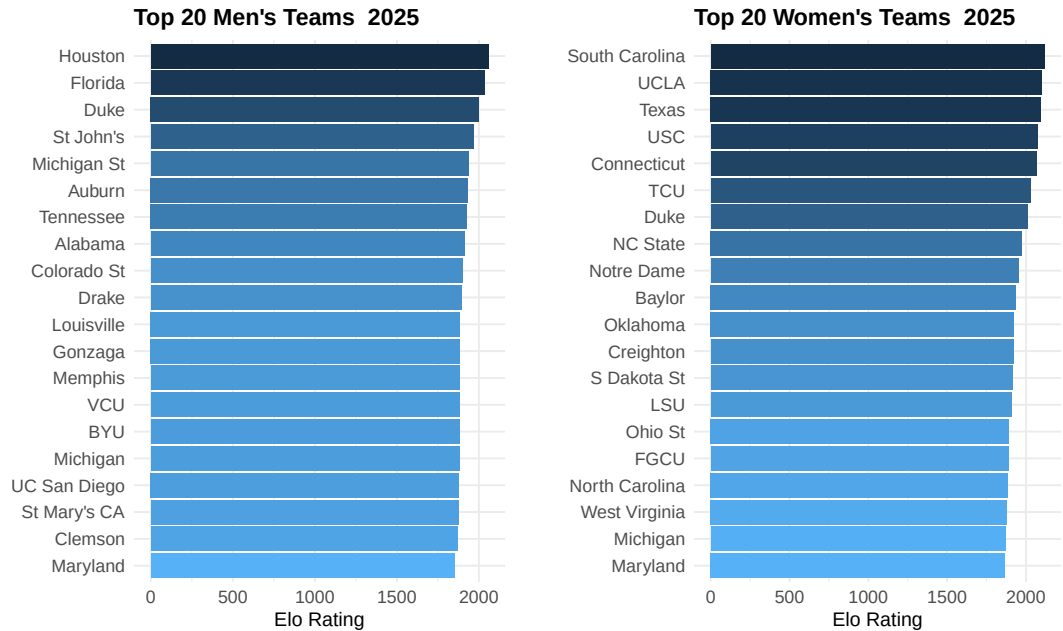


Figure 10. Left: Top 20 men's programs by end of season Elo rating in 2025. Right: Top 20 women's programs by end of season Elo rating in 2025.

Model Implementation

With the training matrix assembled and the feature set finalized, the focus shifts to training the XGBoost model and selecting the configuration that produces the best performance. The model is trained using a binary logistic objective function, which directly optimizes for calibrated win probabilities rather than predicting a continuous point differential that requires a subsequent transformation step. A histogram-based tree method is employed to maximize computational efficiency across the expanded feature space, and a static random seed is set to guarantee reproducibility. Two key

decisions must be made before training can begin: the number of boosting rounds and the hyperparameter values that control the complexity and regularization of each individual tree. Neither can be determined from the training data alone without risking overfitting. To address this, five-fold cross validation, as defined in the Modeling Techniques section, is applied to the full tournament training dataset. At each fold, the model is trained on 80% of the tournament games and evaluated on the held out 20%, with the average loss across all five folds used as the selection criterion. Early stopping is applied with a patience of 50 rounds, meaning training halts automatically if the cross validated performance does not improve for 50 consecutive rounds. Cross validation identified 415 as the optimal number of boosting rounds. Five folds was chosen as a balance between bias and variance in the performance estimate, as fewer folds reduce the training set size while more folds increase computational cost and can produce high variance estimates on small datasets.

In addition to the number of boosting rounds, five hyperparameters were tuned to control the complexity and regularization of the model. Each parameter was tuned independently by fixing all others at their current best value and testing candidate values one at a time, selecting the value that produced the lowest cross validated loss before moving to the next parameter. The learning rate `eta` = 0.01 was selected after testing values of 0.02 and 0.005, producing a smoother and more regularized model at the cost of requiring more boosting rounds to converge. A value of `max_depth` = 5 was selected after testing values of 4 and 6, controlling how many sequential splits

each tree can make. The `subsample = 0.8` parameter means each tree is built on a random 80% of the training data, introducing stochasticity that reduces overfitting. Similarly, `colsample_bytree = 0.7` means each tree randomly samples 70% of the available features, further reducing the influence of any single predictor. Finally, `min_child_weight = 4` sets the minimum number of training observations required in a leaf node before a split is made, preventing highly specific splits that apply to only a handful of games. Predictions for the 2025 tournament are generated by constructing the full feature matrix for every possible matchup using each team's 2025 regular season statistics and final Elo ratings, then applying the fitted model to produce a win probability $\hat{P} \in (0, 1)$ for each pairing.

Model Analysis

The feature importance output from the final model ranks each predictor by its average gain across all splits, where gain measures the reduction in the loss function attributable to each specific feature. As shown in Figure 11, `Seed_diff` is the most influential predictor by a wide margin, accounting for 32.2% of the total gain. This result is entirely consistent with the exploratory analysis presented in the Variable Selection section, which demonstrated a remarkably strong relationship between seed differential and tournament outcomes. Ultimately, this confirms that tournament seeding remains the single strongest signal for predicting matchup results across the historical dataset.

The `Elo_diff` metric ranks second with 11.5% of the total gain, capturing cumulative season long performance in a nuanced way that seeds

alone cannot reflect. Together, these two features account for roughly 44% of the model's total predictive power, forming the core of the algorithm's decision making process. The `avg_point_diff_diff` variable contributes an additional 5.5%, reflecting the average margin by which each team outperformed its opponents during the regular season. Beyond these three primary features, the remaining predictors each contribute less than 3% individually, with strength of schedule variables making up the next tier followed by a long tail of specific box score statistics. This concentration of predictive power in seeding and Elo suggests that broad measures of aggregate team quality drive tournament outcomes significantly more than any individual box score metric.

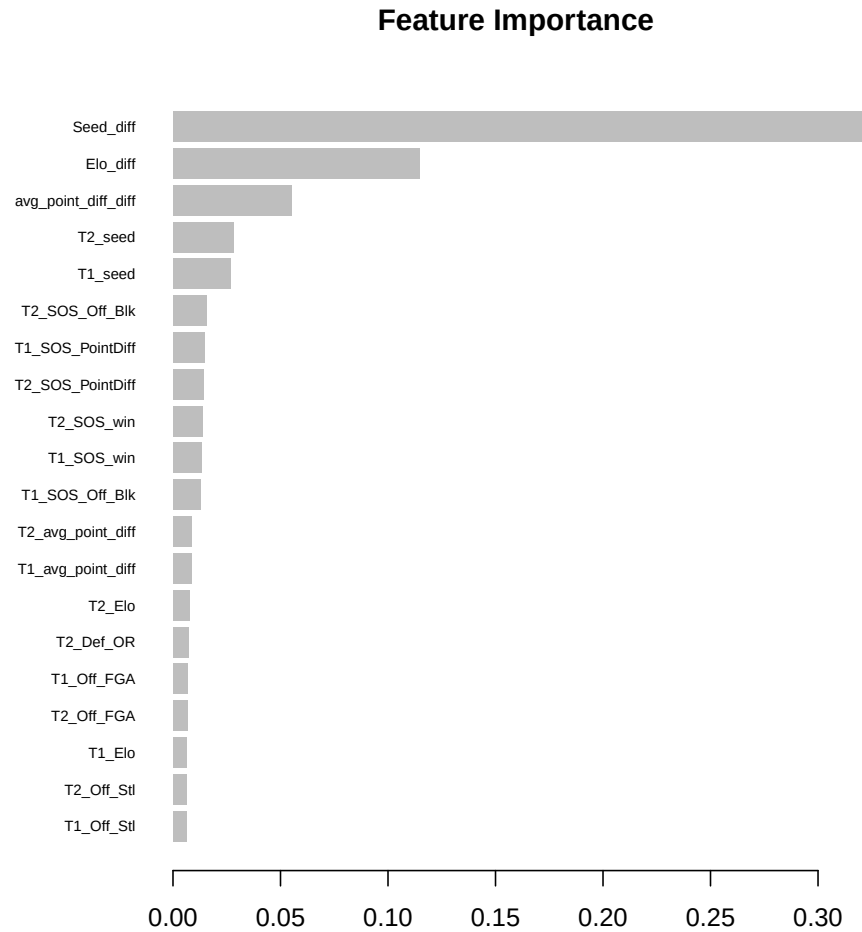


Figure 11. Feature Importance for the XGBoost Model.

The predictive ability of the model is further assessed through the (AUC), presented in Figure 12. The training AUC of 0.9235 confirms that the model possesses strong predictive ability on the training data, meaningfully exceeding the 0.5 baseline that would be expected from a model with no predictive power. This high level of discrimination is consistent with the feature importance findings, where a small number of features like seeding and

Elo dominate the model's capacity to separate wins from losses. Because the AUC is significantly elevated, it provides additional confidence in the model's ability to rank team strength effectively prior to the 2026 deployment.

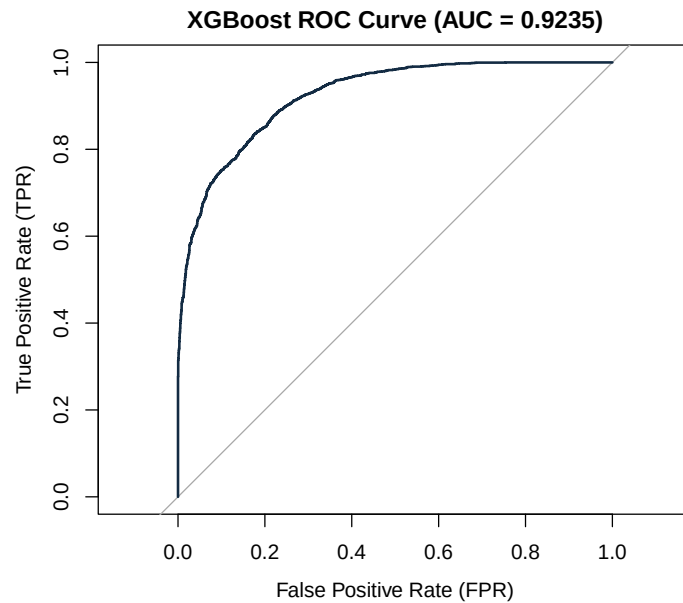


Figure 12. ROC Curve for the XGBoost Model

The logistic regression and XGBoost models developed in the preceding sections predict win probabilities directly from features derived entirely from the Kaggle competition dataset. While both models demonstrated meaningful predictive performance, they share a fundamental limitation: the outcome variable is modeled as a function of aggregated season statistics without any explicit representation of how many points each team is expected to score in a given matchup. The Linear Regression Monte Carlo model takes a different approach by decomposing the prediction problem into

two stages. In the first stage, a linear regression model is trained to predict the number of points a team will score given its own offensive efficiency, its opponent's defensive efficiency, and the pace at which both teams play. In the second stage, the predicted scores and their associated uncertainty are used to simulate each game thousands of times, with the win probability derived as the proportion of simulations in which the team of interest outscores its opponent. This simulation-based framework incorporates external efficiency ratings from two sources: KenPom for the men's division and Bart Torvik for the women's division. Both sources provide opponent-adjusted offensive and defensive efficiency metrics that are not available in the raw Kaggle box score data, allowing the model to draw on a richer and more carefully constructed measure of team quality than the previous two models.

Linear Regression Monte Carlo

The Linear Regression Monte Carlo model is motivated by the observation that predicting a final score is a more tractable regression problem than directly estimating a win probability. Rather than asking the model to learn a complex nonlinear boundary between wins and losses, the linear regression stage asks a simpler question: given two teams' efficiency profiles, how many points should each team score? The residual standard deviation from this regression, denoted σ , captures the inherent unpredictability of college basketball scoring and serves as the engine of the simulation stage. By drawing repeated samples from the implied score distributions and counting the proportion of simulated games in which each team wins, the model

converts a point estimate into a well-calibrated probability that naturally reflects the uncertainty in the underlying score prediction.

Variable Selection

The feature set for the Linear Regression Monte Carlo model is built around opponent-adjusted efficiency ratings sourced externally rather than computed from raw box score totals. For the men’s division, KenPom provides adjusted offensive efficiency (**ORtg**), adjusted defensive efficiency (**DRtg**), and adjusted tempo (**AdjT**), all of which account for the strength of opposition faced and the pace at which games were played [10]. For the women’s division, equivalent metrics are sourced from Bart Torvik, providing adjusted offensive efficiency (**AdjOE**), adjusted defensive efficiency (**AdjDE**), and adjusted tempo (**AdjT**) [3]. These opponent-adjusted metrics are preferred over the raw box score averages available in the Kaggle dataset because they isolate a team’s true efficiency from the confounding influence of schedule strength, providing a cleaner signal for predicting how many points a team will score against a given opponent.

A key design decision in the feature construction is how the efficiency ratings are paired for each prediction. Rather than inputting each team’s statistics in isolation, the model explicitly pits each team’s offensive profile against the opponent’s defensive profile. To predict Team A’s score, the model uses Team A’s adjusted offensive efficiency as **OffRating** and Team B’s adjusted defensive efficiency as **DefRating**, along with both teams’ tempo values. The same logic applies symmetrically to predict Team B’s score. This pairing structure mirrors how scoring actually occurs in basketball — a team’s

expected output is not simply a function of how well it attacks, but how well it attacks against a specific opponent’s defense. A team with an elite offense facing an elite defense should be expected to score fewer points than the same team facing a weak defense, and the feature pairing ensures the model captures this interaction directly rather than leaving it to be inferred from aggregate statistics alone.

Two additional differential features are included to capture relative team quality beyond the efficiency ratings alone. The Elo differential (`Diff_Elo`), computed as the difference between each team’s end-of-season Elo rating, reflects the cumulative outcome of each team’s season relative to its opponents. The Elo rating system used here follows the implementation described in [7], originally written in Python and reimplemented in R for this analysis.

For the men’s division, an additional net rating differential (`Diff_NetRtg`) is also included, defined as the difference between each team’s KenPom net rating, which is the margin between adjusted offensive and defensive efficiency. This feature is not included in the women’s model because the Bart Torvik data source does not provide a precomputed net rating, and computing it manually from `AdjOE` and `AdjDE` was not performed during implementation. This represents a known limitation of the women’s model relative to the men’s, and future iterations should compute and include this feature to fully align the two frameworks. Season identifiers and team identifiers are excluded from the feature matrix as they carry no generalizable predictive signal.

Data Preparation

The data preparation pipeline for the Linear Regression Monte Carlo model differs from the previous two models in two important ways. First, the training data is drawn exclusively from the current regular season rather than historical tournament games. The men’s model is trained on all 2025 regular season games and the women’s model is trained on all 2025 regular season games, meaning the linear regression learns to predict points scored from efficiency ratings and pace as observed in that season’s regular season play. This is appropriate because the external efficiency ratings from KenPom and Bart Torvik are also season-specific, reflecting the adjusted efficiencies accumulated over the 2025 regular season only. Second, the model requires an additional data integration step not present in the previous models: team name matching between the external ratings sources and the Kaggle team identifier system.

Each regular season game is duplicated symmetrically in the same manner as the XGBoost training data, generating one row from the scoring team’s perspective and one from the opponent’s perspective. This ensures the model learns to predict points from both orientations and does not develop a directional bias. The Elo ratings used as a differential feature are computed from the full historical regular season results available in the Kaggle dataset, spanning multiple decades of game results. At the start of the first available season every team is initialized at 1500. Ratings are then updated after every regular season game and carried over between seasons according to $R_{\text{new}} = 0.5 \times R_{\text{final}} + 750$, meaning each team’s end-of-season rating partially

persists into the following year rather than resetting. By the time the 2025 end-of-season ratings are extracted, they reflect the cumulative outcome of over two decades of regular season results, giving historically dominant programs a meaningfully higher baseline than programs with limited historical success. Following symmetric duplication, the training matrix is filtered using `complete.cases` to retain only rows where every feature is present. In practice no rows were removed, indicating that every team in the 2025 regular season had sufficient data across all feature sources to produce a complete profile. The final men’s training matrix consisted of 10,326 rows and 6 features, while the women’s training matrix consisted of 10,828 rows and 5 features, with the women’s model excluding `Diff_NetRtg` as discussed in the Variable Selection section.

Model Implementation

The linear regression model is fit using ordinary least squares, which identifies the coefficient vector $\hat{\beta}$ that minimizes the sum of squared residuals between the predicted and observed scores across all training observations. Unlike the logistic regression described in the prior section, which models the log-odds of a binary outcome, the linear regression here models a continuous response, the number of points a team scores in a given game. The formal specification for the men’s model is:

$$\begin{aligned}
 (15) \quad \text{Points} = & \beta_0 + \beta_1(\text{OffRating}) + \beta_2(\text{DefRating}) + \beta_3(\text{AdjT_team}) \\
 & + \beta_4(\text{AdjT_opp}) + \beta_5(\text{Diff_NetRtg}) + \beta_6(\text{Diff_Elo}) + \varepsilon
 \end{aligned}$$

and for the women's model:

$$(16) \quad \begin{aligned} \text{Points} = & \beta_0 + \beta_1(\text{OffRating}) + \beta_2(\text{DefRating}) + \beta_3(\text{AdjT_team}) \\ & + \beta_4(\text{AdjT_opp}) + \beta_5(\text{Diff_Elo}) + \varepsilon \end{aligned}$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is the residual error term. The residual standard deviation σ is estimated from the training data. The values of σ for both divisions are reported and interpreted in the Model Analysis section, where they serve as a measure of the model's predictive accuracy. Here σ serves as the key input to the simulation stage, quantifying the uncertainty around each predicted score. The resulting coefficients reflect both the direction and magnitude of each predictor's contribution to the expected score. A positive coefficient on **OffRating** indicates that a team with a higher adjusted offensive efficiency is expected to score more points, which is consistent with basketball theory. Similarly, a positive coefficient on **DefRating** reflects the opponent's defensive efficiency, a higher opponent defensive rating means the opponent allows more points per possession, which naturally increases the scoring team's expected output. A positive coefficient on **AdjT_team** indicates that teams playing at a faster pace are expected to score more points in absolute terms, since more possessions create more scoring opportunities, while **AdjT_opp** captures the opponent's contribution to the pace of the game.

To convert the predicted scores into win probabilities, a Monte Carlo simulation is applied to each tournament matchup. For a given matchup between Team A and Team B, the linear regression produces two point estimates $\hat{\mu}_A$ and $\hat{\mu}_B$, representing the expected scores for each team given

their respective efficiency profiles and the matchup context. Score realizations are then drawn independently for each team:

$$(17) \quad \text{Score}_A \sim \mathcal{N}(\hat{\mu}_A, \sigma^2)$$

$$(18) \quad \text{Score}_B \sim \mathcal{N}(\hat{\mu}_B, \sigma^2).$$

The win probability for Team A is then computed as:

$$(19) \quad \hat{P}(\text{Team A wins}) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{Score}_{A,i} > \text{Score}_{B,i}]$$

This proportion $\hat{P}(\text{Team A wins})$ is submitted directly to Kaggle as the predicted win probability for the matchup, with the complementary probability $1 - \hat{P}(\text{Team A wins})$ implicitly assigned to Team B. The normality assumption for score distributions is motivated by the central limit theorem: a team’s final score is the aggregate of many independent possession outcomes, and the distribution of these aggregates converges toward normality as the number of possessions increases. The men’s model uses $n = 100,000$ simulations per matchup, and the women’s 2025 submission used $n = 50,000$ due to an implementation inconsistency discussed in the limitations section. At both sample sizes, the standard error of the proportion estimate is bounded by $\sqrt{p(1-p)/n} \leq 0.002$ at $n = 50,000$ in the worst case, confirming that the estimates have converged to stable values.

As with the logistic regression, separate models are trained for the men's and women's divisions rather than a single combined model. The justification mirrors that of the prior section. The relationship between efficiency ratings, tempo, and scoring may differ structurally between divisions due to differences in pace, physicality, and playing style. Training separate models allows each set of coefficients to reflect the unique dynamics of its respective division rather than forcing shared parameter estimates that could mask meaningful differences.

An important practical consideration in this implementation is the decision to train exclusively on the current regular season rather than pooling multiple years of data. A single NCAA regular season produces several thousand games per division, which is sufficient to estimate a low-dimensional model with six or seven predictors reliably. However, this also means the model is sensitive to season-specific variation in how efficiency ratings translate to scoring. A team whose KenPom ratings were inflated by a weak schedule may be overestimated by the model, and conversely, a team that peaked late in the season after an early slump may have efficiency ratings that understate their true tournament form. These limitations are inherent to any model that relies on season-aggregate statistics and are shared to varying degrees by all three modeling approaches in this analysis. For this reason, the model is deliberately kept parsimonious, relying on six predictors for the men's model and five for the women's rather than expanding the feature set further. This is consistent with the principle of Occam's Razor applied in the logistic regression section. A simpler model that generalizes reliably is

preferred over a more complex one that risks overfitting to season-specific patterns in the training data [23].

Model Analysis

With the models estimated on the 2025 regular season data, the resulting coefficients can be examined to evaluate which predictors carry meaningful weight in each division and how well the linear regression captures the relationship between efficiency ratings and scoring. Because separate models were trained for the men’s and women’s tournaments, the coefficient structures can be analyzed independently and then compared to assess whether the efficiency and tempo features operate differently across divisions. Tables 7 and 8 present the estimated coefficients, p-values, and significance levels for the men’s and women’s models respectively.

Table 7. Linear Regression Coefficient Estimates (Men’s 2025 Model)

Variable	Estimate	p-value	Sig.
Intercept	-207.3986	0.0000	***
Off Rating	0.6519	0.0000	***
Def Rating	0.6421	0.0000	***
AdjT Team	1.0826	0.0000	***
AdjT Opp	1.0086	0.0000	***
Diff NetRtg	0.0540	0.0029	**
Diff Elo	0.0013	0.1229	

In the men’s model, four of the six predictors reach conventional statistical significance at the 0.001 level: `OffRating`, `DefRating`, `AdjT_team`, and `AdjT_opp`. The positive coefficient on `OffRating` ($\hat{\beta} = 0.6519$) confirms that teams with higher adjusted offensive efficiency are expected to score more points, consistent with the feature’s definition. The positive coefficient

Table 8. Linear Regression Coefficient Estimates (Women's 2025 Model)

Variable	Estimate	p-value	Sig.
Intercept	-202.9007	0.0000	***
Off Rating	0.7309	0.0000	***
Def Rating	0.7866	0.0000	***
AdjT Team	0.9605	0.0000	***
AdjT Opp	0.8542	0.0000	***
Diff Elo	0.0024	0.0000	***

on **DefRating** ($\hat{\beta} = 0.6421$) reflects that facing a weaker defense, one that allows more points per 100 possessions, is associated with higher expected scoring. Both tempo features carry positive and similarly sized coefficients, with **AdjT_team** ($\hat{\beta} = 1.083$) and **AdjT_opp** ($\hat{\beta} = 1.009$) indicating that faster pace from either team is associated with more points scored in absolute terms, consistent with the interpretation that more possessions generate more scoring opportunities. **Diff_NetRtg** reaches significance at the 0.01 level ($p = 0.0029$), contributing a small but meaningful positive adjustment to the expected score when a team's net rating differential over its opponent is favorable. **Diff_Elo**, however, fails to reach significance ($p = 0.1229$), suggesting that once the KenPom efficiency ratings and net rating differential are accounted for, the Elo differential contributes no additional explanatory power in the men's model.

The women's model tells a somewhat different story. All five predictors reach statistical significance, including **Diff_Elo** ($\hat{\beta} = 0.002383$, $p < 0.001$), which was insignificant in the men's model. The efficiency and tempo coefficients are larger in magnitude for the women's model than the men's, with **OffRating** at 0.7309 versus 0.6519 and **DefRating** at 0.7866 versus

0.6421, suggesting that adjusted efficiency ratings are more predictive of actual scoring in the women's game. The significance of `Diff_Elo` in the women's model but not the men's is directly attributable to the absence of `Diff_NetRtg` in the women's feature set. In the men's model, `Diff_NetRtg` and `Diff_Elo` share variance since both capture overall team quality differential, and `Diff_NetRtg` absorbs the signal that `Diff_Elo` would otherwise contribute. In the women's model, with `Diff_NetRtg` absent, `Diff_Elo` becomes the primary carrier of that signal and its contribution is clearly detectable.

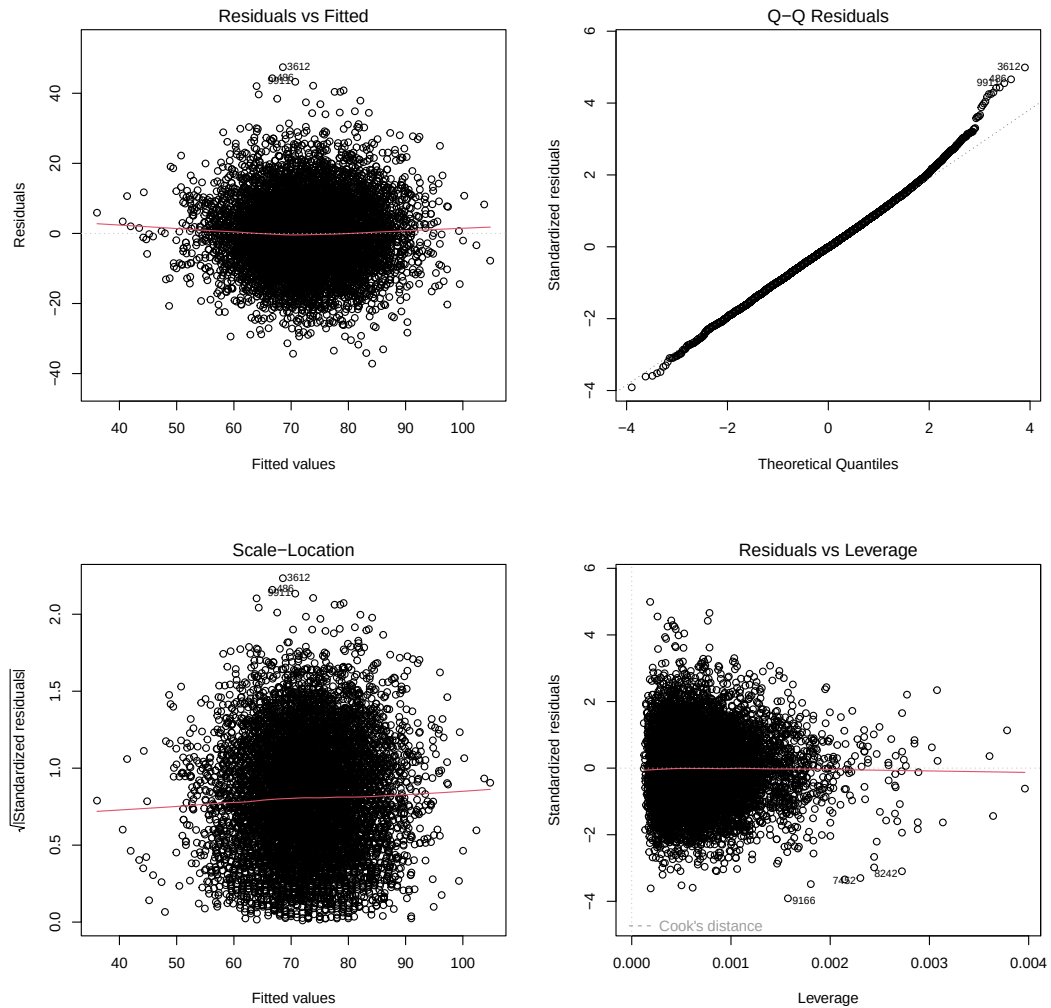


Figure 13. Diagnostic Plots for the Men's Linear Regression Model

The two models also differ meaningfully in their goodness of fit. The women's model achieves an R^2 of 0.524, meaning it explains 52.4% of the variance in points scored, compared to 0.379 for the men's model. This gap suggests that Bart Torvik efficiency ratings are more predictive of women's scoring outcomes than KenPom men's rankings. This is possibly because the women's game exhibits less of the high variance play that characterizes men's

basketball at the tournament level [33]. The residual standard deviations are $\sigma = 9.507$ for the men's model and $\sigma = 9.300$ for the women's model, meaning the women's model's score predictions are marginally more precise. These values of σ feed directly into the Monte Carlo simulation as the spread of the score distribution, so the women's slightly lower σ means its win probability estimates are derived from slightly tighter score distributions.

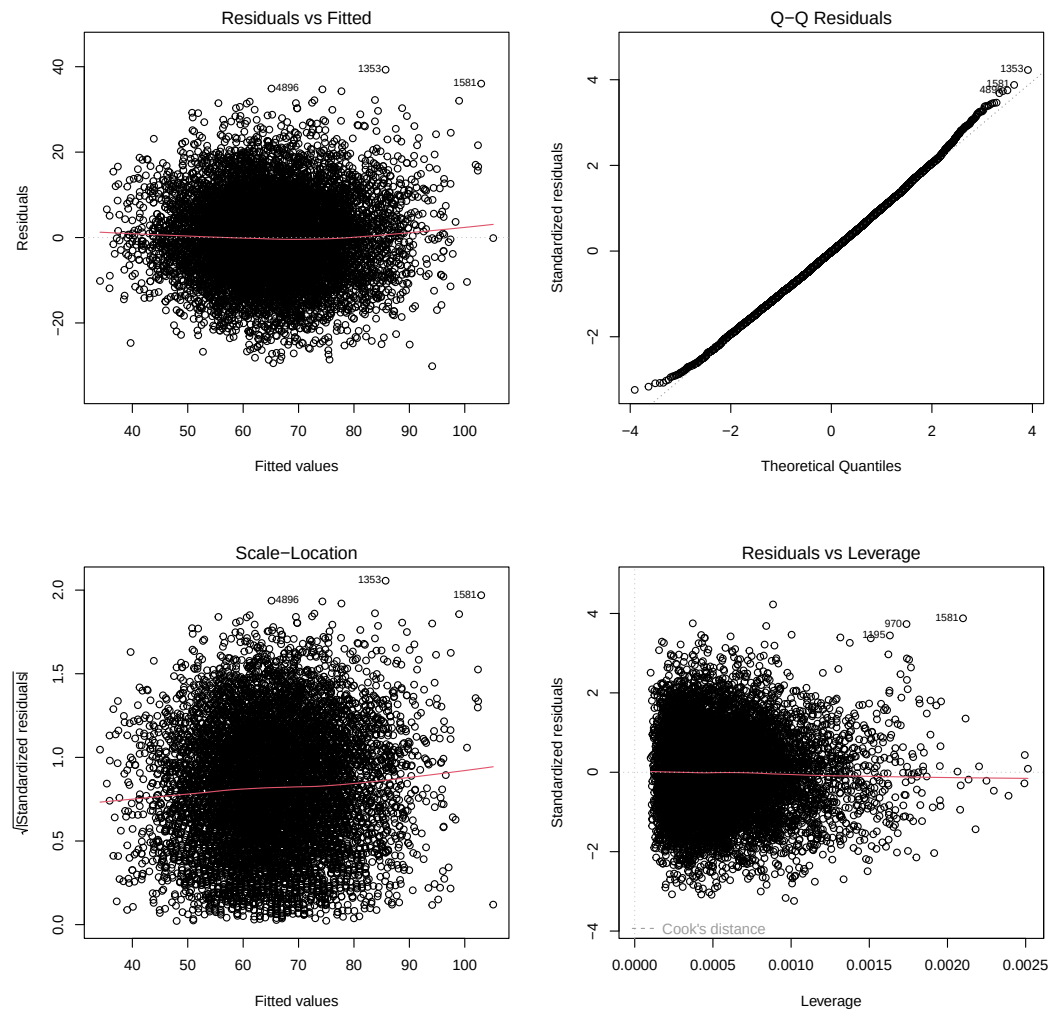


Figure 14. Diagnostic Plots for the Women's Linear Regression Model

Figure 13 presents the diagnostic plots for the men’s model and Figure 14 presents the equivalent plots for the women’s model. In both cases, the Residuals vs. Fitted plots show residuals randomly scattered around zero with approximately flat smoothing lines, confirming that the linearity assumption is reasonably satisfied across the range of predicted scores. The Q-Q plots show that the residuals follow the theoretical normal line closely through the central range in both models, with some deviation in the tails where a small number of extreme scoring observations pull away from the reference line. This tail behavior is driven by the frequency of extreme blowouts that act as outlying observations, a phenomenon well-documented in empirical sports modeling [34]. This deviation does not materially undermine the normality assumption that motivates the simulation framework, particularly given the large training sample sizes of over 10,000 observations in each division. The Residuals vs. Leverage plots confirm that no observations exert undue influence on the model estimates in either division, as no points fall outside Cook’s distance boundaries and leverage values remain uniformly small.

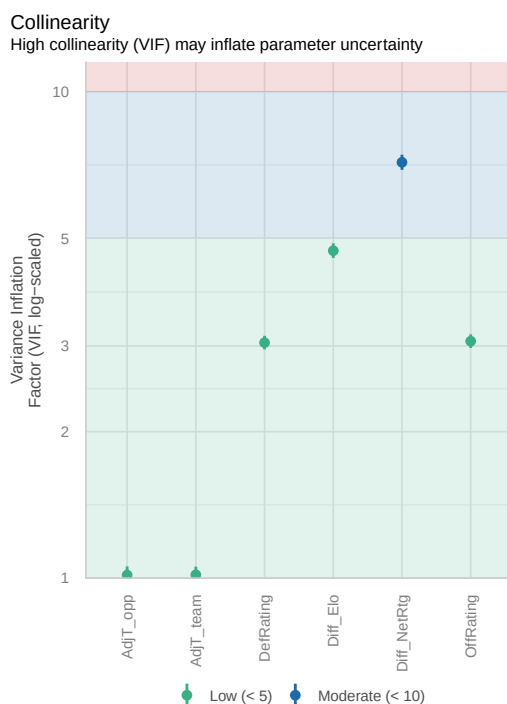


Figure 15. VIF Plot for the Men's Linear Regression Model

The VIF analysis reveals a meaningful structural difference between the two models. In the men's model, shown in Figure 15, `Diff_NetRtg` carries a moderate VIF of approximately 7.5, the highest among all predictors, indicating that it shares substantial variance with other features in the model. This collinearity is the direct cause of `Diff_Elo`'s insignificance in the men's model, as the two features capture overlapping information about team quality differential. In the women's model, shown in Figure 16, with `Diff_NetRtg` absent, all VIF values fall below 5 and are classified as low, indicating a cleaner feature structure with no problematic collinearity. While the men's model's moderate collinearity does not invalidate the estimates, it inflates the standard errors of the affected coefficients and makes it more

difficult to isolate the individual contributions of correlated predictors. This is an accepted consequence given that both features were included for their theoretical relevance rather than purely statistical grounds.

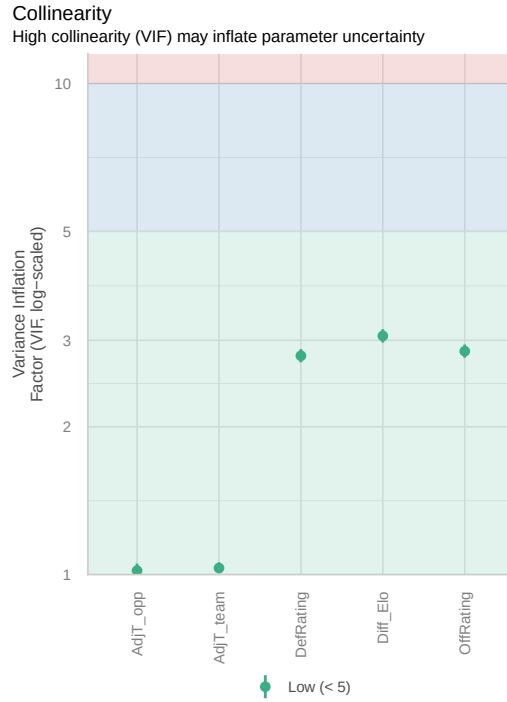


Figure 16. VIF Plot for the Women's Linear Regression Model

Taken together, the diagnostic analysis confirms that both models satisfy the core assumptions of linear regression sufficiently to justify their use in the simulation framework. The women's model is the stronger of the two by every measured criterion, achieving higher R^2 , lower σ , cleaner VIF structure, and full predictor significance. The men's model is adequate but limited by the moderate collinearity between `Diff_NetRtg` and `Diff_Elo` and the lower overall explanatory power of the efficiency features in the men's regular season data. These findings reinforce the decision to train the models

independently rather than pooling the divisions, as the structural differences in coefficient magnitudes, predictor significance, and goodness of fit confirm that the two divisions are not governed by the same scoring dynamics.

RESULTS

Model Comparison

2026 Competition

talking about the results of each model in the competition, as well as our submission.

CONCLUSIONS

i am free

Summary

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